

Kuiyi (Cory) Gao

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Mandarin (native) | English — TOEFL iBT 110 | Cantonese (intermediate)

Education

University of North Carolina at Chapel Hill

Aug 2026 – May 2027

B.A. in Computer Science (transfer); exchange student Jan – May 2026

(expected)

- Cumulative GPA 3.68/4.00.
- Advanced coursework: COMP 550 Natural Language Processing, COMP 562 Machine Learning, COMP 586 Advanced Natural Language Processing (research project), COMP 380 Technology, Ethics & Culture.

The Chinese University of Hong Kong

Sep 2023 – May 2026

B.Sc. in Computer Science, Faculty of Engineering ELITE Stream

- Advanced coursework: ELEG 5491 Introduction to Deep Learning (graduate level), ESTR 3108 Fundamentals of Artificial Intelligence (ELITE stream), CSCI 3130 Formal Languages & Automata Theory, CENG 3420 Computer Organization & Design, CSCI 2100 Data Structures, ENGG 2760/2780 Probability & Statistics for Engineers, Applied Deep Learning, Operating Systems, Database Design & Analysis, Linear Algebra.

Publications

- [1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. **ACL 2025 (Main Conference)**. aclanthology.org/2025.acl-long.832
- [2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. **Findings of ACL 2025**. aclanthology.org/2025.findings-acl.571
- [3] **Kuiyi Gao*** et al. *A cross-platform benchmark for GUI-agent robustness under interface noise*. Under review, 2026 (*co-first author).

Research Experience

GUI-Agent Robustness under Interface Noise — CUHK, advised by Prof. Michael R. Lyu

Hong Kong & remote
Sep 2025 – present

- Co-first author of a dynamic, cross-platform (web / desktop / mobile) benchmark: 42 interface-noise types, 7 agent frameworks evaluated in live executable environments against task-level rubrics; noise redirects trajectories and raises unsafe-action rates, not only failure rates.
- Designed the evaluation metric suite (pass@k, relative degradation, action-difference rate, unsafe-action rate, correct-prefix length) and the mock-interface harness for computer-use, browser-use and self-operating agents.

Chain-of-Jailbreak: Editing-Step Attacks on T2I Models — Tencent AI Lab & CUHK; with Michael R. Lyu, Wenxuan Wang, Zhaopeng Tu

Hong Kong
Jun 2024 – Feb 2025

- Designed the attack methodology (step-by-step editing that decomposes a harmful query into benign edits); built part of the dataset; ran most attack and defense experiments; wrote the paper and led the rebuttal. Second author, Findings of ACL 2025 [2].

MMSafeAware: Multimodal Safety Awareness — Tencent AI Lab; with Zhaopeng Tu, Wenxuan Wang

Hong Kong & Shenzhen
Dec 2023 – Oct 2024

- Designed the evaluation methodology, built the benchmark (1.5k image-prompt pairs across 29 safety scenarios, covering both unsafe and over-sensitive responses) and ran the vision-centric reasoning fine-tuning experiments. Third author, ACL 2025 main conference [1].

Projects

Agent Skill Firewall — MBZUAI UGRIP; with Dr. Steve Liu

Abu Dhabi & Beijing
(hybrid)
Jun – Jul 2026

- **Built** the runtime egress firewall for agent skills in a four-student team: a static pre-scan of the bundle, then intent–action–result verification of live traffic through an HTTP proxy seeded with canary tokens.
- **Shipped** it as a usable tool: TUI and web console, Docker deployment, 23 unit tests, CI across Linux and macOS on Python 3.10–3.12.
- **Measured** the question the tool raised: on stored runs of 65 malicious skills the detector flagged none, because 46 of the 65 produced no proxy-visible egress. Reported recall therefore factors as $P(\text{execute}) \cdot P(\text{observe} | \text{execute}) \cdot P(\text{detect} | \text{observed})$.
- *Status: system complete; measurement study in preparation. Tools: Python, HTTP interception, Docker, GitHub Actions.* [\[code\]](#)

Kernel-Based Novelty Scores for Text — CUHK summer research; with Prof. Farzan Farnia

Hong Kong
May – Aug 2025

- **Extended** the Kernel Entropic Novelty (KEN) score from images to text — whitening for isotropy, plus a differentiable Nyström approximation — and **formulated** a personality-conditioned multi-objective alignment criterion on top of it. *Status: exploratory, proposal stage.*

Memory × Persona Safety — a pre-registered research program

Independent
Jul 2026 – present

- **Pre-registered** five studies (steering dose law, retrieval-induced forgetting, memory confidence without metadata, memory-mediated sycophancy drift, deletion-influence residual audit) behind a shared experiment harness with 28 smoke tests. *Status: proposal stage.*
- **MemSlot saliency + DeepSeek-OCR compression for long contracts** (UNC COMP 586, 2026): a frozen encoder with 32 learnable memory slots learns which spans of a contract matter before a question is asked; those spans are then rendered and compressed through an OCR model. [\[code\]](#)
- **Course AI teaching assistant** (CUHK GEWS2011, 2025): Flask web app whose two-stage LLM pipeline drafts an outline under the course's academic-integrity guideline before answering — guidance rather than answers; invitation-code accounts, admin console, Render deployment.
- **Objectivity of Imagist poetry measured with text-to-image models** (CUHK ENGE3940, 2025): 20 images per poem from Flux.1-dev and SD3, CLIP-embedding variance as the statistic — Imagism 0.12 vs. Romanticism 0.28 (Flux, $p = 0.0499$).
- **Movability segmentation for embodied safety** (CUHK ELEG 5491, 2025): DeepLabV3 with a custom ASPP head on COCO-Stuff relabelled into four movability levels; mIoU 0.638 at 58 FPS. [\[code\]](#)

Honors & Awards

- CUHK Faculty of Engineering ELITE Stream; Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).
- Gaokao (national college entrance examination): top 2% in Beijing, 2023.

Skills

- **Languages:** Python, C, Java, JavaScript/HTML/CSS, $\text{\LaTeX}$. **Frameworks & tools:** PyTorch, Hugging Face Transformers, Flask, Docker, GitHub Actions CI, Playwright / GUI-agent harnesses, LLM APIs (OpenAI, Anthropic, DeepSeek).